



Breathing with lungs

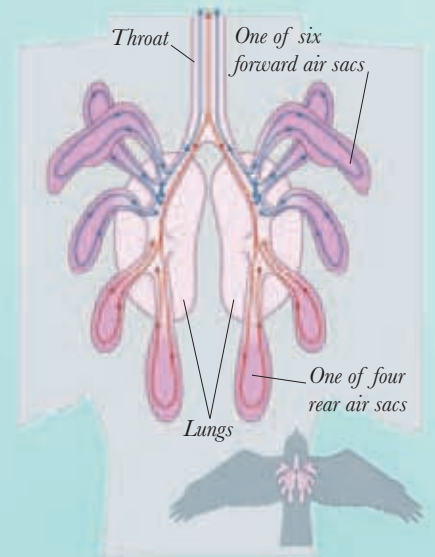
On a cold winter's day, a red deer produces a cloud of mist as it roars to rival males. Like all mammals, it breathes with a pair of lungs, which are hollow spaces on either side of the heart. To breathe in, it uses muscles that expand its chest, sucking in fresh air. When the breathing muscles relax, the air moves back out. To roar, the deer takes an extra-large breath and then forces it through its throat.

One-way traffic

When mammals breathe, air travels into their lungs and then back out again. A bird's lungs are much better at collecting oxygen, because air flows straight through them, instead of in and out. Air sacs connected to the lungs keep the air on the move. Thanks to this super-efficient system, birds can breathe high in the sky, where the air is very thin.

Air flow

When a bird breathes in, oxygen-rich air (shown in red) moves into its rear air sacs. After flowing through the lungs, stale air (shown in blue) moves into the forward air sacs, before being breathed out.



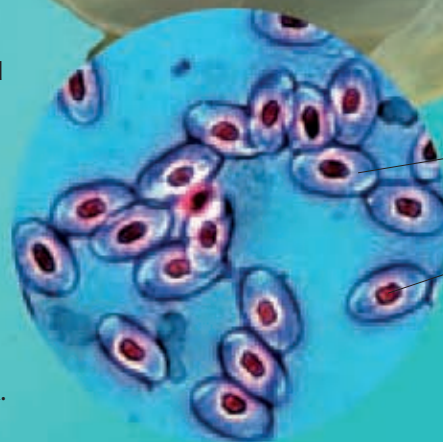
Red color produced by hemoglobin

Network of blood vessels carries blood around frog's body



Vital fluid

Pumped by the heart, blood is like a liquid conveyor belt that moves oxygen and nutrients around an animal's body. In most vertebrates, including this glass frog, oxygen is carried by hemoglobin—an iron-containing protein stored inside red blood cells. Blood circulates many other crucial substances around the body. Among them are glucose, the body's main source of energy, and hormones, which are chemical messengers that affect the way cells work.



Oval-shaped red blood cell

Nucleus, or control center, containing DNA (genetic instructions for the cell)

Microscopic view of frog blood